

CEBE Interdisciplinary Fellowship Programme
Call for PhD and PostDoc Projects

Human–AI Collaborative Workflows for Immersive Digital Twins in Bridge Monitoring and Inspection (HAI-BRI)

PhD project

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1 Project Summary

This PhD project **HAI-BRI** will develop human–AI interaction framework for bridge Structural Health Monitoring (SHM). The framework will integrate Unmanned Aerial Systems (UAS) imagery, sensor measurements, structural information, inspection histories, environmental observations, and engineering knowledge through a Digital Twin (DT). Evidence will be semantically contextualized into an evolving representation of the bridge condition described as a Knowledge Graph (KG). Artificial Intelligence (AI) based on Large Language Models (LLM) and Vision-Language Models (VLM) will support interpretation of this evidence by identifying relevant relationships, plausible deterioration mechanisms, uncertainty, and information gaps. Extended Reality (XR) will be explored to create an immersive environment for the users.

The framework will be validated experimentally by comparing conventional engineering workflows, AI assistance without semantic grounding, and the proposed knowledge-grounded human–AI workflow. The framework will be validated on two bridge case studies in the Global South, namely, the Ponte do Funil (Itaparica Island, Bahia, Brazil), supported by the Federal University of Bahia, and the elevated viaduct sections of the Nairobi Expressway (Nairobi, Kenya), supported by the University of Nairobi. Deterioration mechanisms associated with water ingress, expansion joints, drainage deficiencies, moisture, cracking, and corrosion-related symptoms will be prioritized.

2 Project Description

2.1 Motivation, Significance and Scientific Challenges

Deterioration of bridge infrastructure is a critical issue in Western countries, where such assets, in large part, already exceed their nominal life. The catastrophic collapse of the Morandi Bridge (Genova, Italy, 2018) indicates that bridges in poor condition may still be operated. The Global South is characterized by younger infrastructure; however, underfunded maintenance programs are responsible for faster deterioration. Vibration-based SHM has been implemented extensively in the last 20 years along with the large diffusion of cost-effective sensing technology [1]. More recently, UASs have become affordable and easy to operate, facilitating the capture of high-resolution image data from hard-to-access structural elements [2]. However, engineers must still combine these heterogeneous sources to determine whether an observed condition is significant, whether additional inspection is needed, and which intervention to prioritize [3]. The overwhelming amount of data, along with the shortage of qualified personnel, renders this a challenging task.

The project’s vision is to develop a human-centered DT that supports engineers in designing and analyzing inspection campaigns. AI provides the opportunity for engineers to rapidly access and interpret a variety of data sources to support them in their decision-making process. However, effective human–AI collaboration workflows are largely underexplored for inspection and SHM workflows. Moreover, the integrating of immersive technologies to enhance user experience and decision-making for inspection workflows remains largely unexamined.

The research addresses this gap in three ways: 1) by a building a system in which AI reasoning is grounded in a semantic DT and delivered through Extended Reality (XR); 2) by developing and evaluating human–AI collaboration workflows for immersive bridge inspection; and 3) by providing an empirically grounded framework for future inspection workflows involving expected expert users.

The PhD project funded by **HAI-BRI** will address the two following research questions:

- Q1 how should the semantic representation of a bridge evolving condition be structured and integrated with AI so that reasoning on its condition is logically consistent and traceable to empirical evidence?
- Q2 how should the DT operator interact with the AI, through conversational and immersive interfaces, to build situational awareness and make better-supported inspection and maintenance decisions?

Aarhus University (AU) will tackle Q1 while ETH Zurich (ETHZ) will tackle Q2. The key-enabling technologies are 1) semantic representations of infrastructure condition, 2) multimodal LLM/VLM reasoning grounded in the DT, and 3) XR providing the immersive layer.

Q1 will be studied by comparing levels of semantic granularity and AI reasoning with and without graph-based grounding, in terms of evidence retrieval, consistency, traceability, and unsupported conclusions; Q2 by user studies and controlled experiments comparing conventional, conversational, and immersive access to the DT on decision performance, SA, cognitive workload, trust and reliance on AI.

The PhD Candidate (Lorenzo Loyola) is educated as a structural engineer and has +5 years of experience in R&D (academia and industry) related to the digitalization of structural design and monitoring workflows. He is already employed at AU. So the project will start in January 1st, 2027 and last 36m.

2.2 State of the Art

A DT combines digital and physical assets into a cyber-physical system to deliver specific services [4]. In the context of SHM, DTs are used to collect multi-modal data related to a structural system, process it, and inform decision making [5]. As of today, machine learning is used to automate some specific processing tasks (e.g., crack localization) [3]. However, decision making regarding inspection and maintenance still requires expert users to reason across large data sets, considering the entire history of the asset [6]. The SHM community recognizes that such an approach leaves substantial information unexploited and leads to sub-optimal engineering decisions. Recent work points toward the use of AI to overcome this limit [7].

Ontologies are being explored to add a semantic description to DT components and functionalities and facilitate user interaction. An ontology provides a formal, machine-interpretable representation of domain concepts and their relationships [8]. DTs, therefore, increasingly employ ontologies as semantic layers for interoperability, traceability, and machine reasoning [9, 10]. Ontologies have been recently investigated for turning fragmented inspection and maintenance data into structured knowledge for condition assessment of bridges [11, 12] with the goal of increasing Situational Awareness (SA). In fact, engineers must not only access heterogeneous information but also perceive changes in structural condition, understand their implications, and anticipate their evolution [13]. In parallel, conversational agents and ontologies are explored to operate a DT in place of a graphical user interface [12].

XR, which comprises immersive technologies such as Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), shows great promise for enhancing DTs, both as a gateway to experiencing data that simulates or extends the real world [14] and as a means of improving usability [15], collaboration, and decision-making [16]. XR is also being explored as a way to enhance the manual bridge inspection process independently of DTs [17, 18]. Immersive DT environments have been combined with interactive virtual-agent interfaces for infrastructure leak simulation and monitoring [19].

Nevertheless, important gaps remain. The outlined state of the art is fragmented, single-technology-centered, and engineers SA is overlooked [6]. Human involvement is limited to

validating AI outputs, refining generated knowledge, or facilitating interaction with digital information [7, 12], rather than being explicitly modeled as part of a joint human–AI reasoning and decision-making system.

2.3 Scientific Approach, Methodology, and Novelty

The project will develop and experimentally validate a human-centered framework for bridge SHM in which AI and XR increase operator awareness of the bridge condition. Two bridge case-studies, namely, the Ponte do Funil (Itaparica Island, Bahia, Brazil) and the elevated viaduct sections of the Nairobi Expressway (Nairobi, Kenya), are provided by the Federal University of Bahia School of Engineering (FUBSE) and the University of Nairobi (UN), respectively (Fig. 1). Both are concrete structures exposed to material deterioration (cracks and corrosion). A central research challenge will be preventing the AI from presenting unsupported conclusions

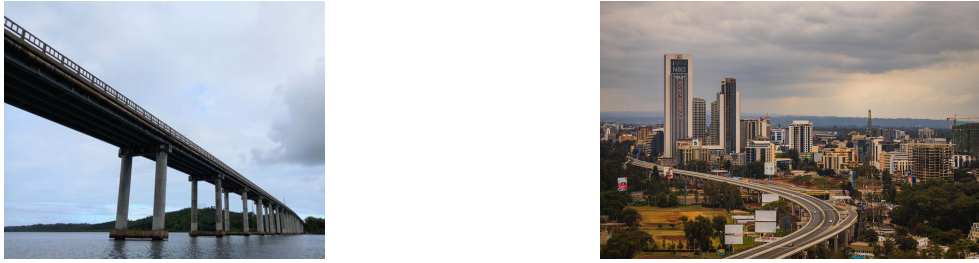


Figure 1: **HAI-BRI** case studies: Ponte do Funil, Brazil (left) and Nairobi Expressway, Kenya (right).

as established facts. A semantic description of the evolving bridge condition will constrain and contextualize AI reasoning, providing engineering-sound assessments of the bridge state and evidence for decision support. For example, when moisture is detected below an expansion joint, the AI retrieves the joint condition and inspection history, drainage configuration, recent rainfall exposure, nearby cracking or corrosion observations, and relevant engineering knowledge before proposing plausible deterioration hypotheses. In this project, the DT is an evolving representation that integrates BIM/FEM models with inspection observations, sensor measurements, environmental information, historical records, deterioration states, and maintenance information. The semantic layer that connects these heterogeneous representations is implemented as a KG. This will allow the DT to evolve as new evidence becomes available and provide a machine-interpretable representation of the infrastructure situation. The research program of **HAI-BRI** is organized into three interconnected Work Packages (WPs). **WP1 Asset knowledge representation (m1-m12)**: The main scientific contribution of WP1 will be a semantic model that links all digital assets associated with the bridge case-studies:

T1.1 Digital Twin foundation: Technical drawings, historical inspection and maintenance records, monitoring data, and other available information on the bridge case-studies will be collected and structured. A BIM model will represent geometry and structural components, while a FEM model will represent mechanical behavior. Experimental data will be organized in databases.

T1.2 Semantic layer: A semantic layer will connect the BIM/FEM models with sensors, measurements (e.g., UAS imagery, inspection records), deterioration mechanisms (e.g., cracking and corrosion), and maintenance plans. Ontologies will define the relevant concepts and relationships, which will be instantiated as a KG.

WP2 Multimodal and Graph-Grounded AI (m6-m24): The main scientific contribution of WP2 will be a multimodal, graph-grounded AI architecture for traceable infrastructure reasoning for evidence-based interpretation of bridge condition:

T2.1 Automated annotation of deterioration: WP2 will investigate how AI can transform heterogeneous observations into evidence-grounded interpretations of bridge condition. LLM/VLM will identify and contextualize candidate observations from data. The objective is not to develop another universal defect detector, but to determine how automatically extracted observations can become structured evidence within the semantic DT.

T2.2 Automated reasoning on structure condition: A graph-grounded LLM/VLM reasoning architecture will retrieve relevant asset-specific information from the KG before generating an engineering interpretation. The quality of AI-supported assessment changes as different information sources are removed or become incomplete will also be investigated.

WP3 Development and Evaluation of an Immersive Human–AI Collaboration Workflows Framework (m18-m36): WP3 will develop and evaluate methods of human–AI collaborative workflows that support trust, appropriate reliance, and decision-making for bridge inspection within immersive environments:

T3.1 Human–AI Collaboration Workflows: T3.1 develops and implements the conversational agent connected to the DT, building the basis of the human–AI workflow. It will associate physical bridge components with current observations, historical changes, and sensor information that the user can query to build SA.

T3.2 Immersive Component (XR): T3.2. will enable immersive processes for experiencing the DT. Together with stakeholders, XR will be explored for their use with DTs for on-site and remote use, and paired with the workflow presented in T3.1.

T3.3 Experimental Evaluation: The framework will be evaluated in laboratory studies with realistic bridge assessment scenarios and in the field. Experienced participants will perform representative tasks (e.g., identification of deterioration mechanisms, planning of inspections) with conventional engineering workflows, with AI assistance without semantic grounding, and the proposed knowledge-grounded human–AI workflow.

Developed technology impact: Faster interpretation of inspection information, less effort searching fragmented records, better prioritization of inspection resources, and traceable evidence supporting engineering decisions.

2.4 Research Environment and Supervision

The PhD project is hosted at AU, with Assoc. Prof. Giuseppe Abbiati as main supervisor. The PhD Candidate (Lorenzo Loyola) will spend approximately 18 months at ETHZ.

Assoc. Prof. Giuseppe Abbiati (AU) has expertise in DT engineering for SHM and will support the implementation of the DTs for the bridge case-studies.

Assist. Prof. Jhonattan G. Martinez (AU) has expertise in automated inspections and will support the formulation of the semantic representation and its integration with LLMs.

Prof. Christos Georgakis (AU) has expertise in bridge engineering and SHM and will support the formulation of the inspection plans for the bridge case-studies.

Dr. Carolin Reichherzer (ETHZ) has expertise in human-computer interaction in immersive environments. She will support development of human-centred XR immersive environments.

Prof. Ueli Angst (ETHZ) has expertise in material durability and will supervise the formulation of the reasoning chain for deterioration assessment for the bridge case-studies.

2.5 Stakeholders involved in the project

Besides the project partners (AU and ETHZ), **HAI-BRI** involves the following stakeholders, both situated in the Global South:

- **Prof. Dayana Bastos, Federal University of Bahia School of Engineering (FUBSE), Department of Structural and Construction Engineering, Brazil:** structural engineering and SHM expertise; access to the Ponte do Funil.
- **Prof. Peter Njeru Njue, University of Nairobi (UN), Department of Real Estate, Construction Management and Quantity Surveying, Kenya:** geomatics and construction management expertise; access to the elevated viaducts of the Nairobi Expressway.

3 Project relevance to CEBE research fields

RF7 – Extending the life of the built environment (primary). **HAI-BRI** addresses the RF7 mission of “decision support that integrates structural health monitoring information” [20, p. 132]. It builds on the AI/robot-assisted operation of SHM systems [20, p. 148] by adding the layer that turns their observations into condition hypotheses linked to deterioration mechanisms such as rebar corrosion. It contributes to practical guidance for risk-informed decisions on repair, retrofitting, and transformation [20, p. 145] by quantifying how engineers rely on AI-supported assessments. It addresses the limited confidence in decisions based on scarce monitoring data and the adoption of new digital tools by practitioners [20, p. 141].

RF4 – Digitalization and automation. The semantic DT and KG of WP1 extend the semantic modeling, reality capture, and interoperability of WP4.4–4.6 [20, pp. 79–80] from buildings to in-service bridge infrastructure. The project will contribute to the RF4/RF7 knowledge exchange on DT engineering and agentic AI [20, p. 146].

RF5 – Climate resilient and adaptive infrastructure. Water ingress is driven by precipitation and exposure. The KG will represent environmental exposure as context for interpreting moisture and corrosion symptoms, interfacing with RF5 work on climate and infrastructure data (WP5.1)[20, p. 102].

RF1 – Assessing and measuring environmental sustainability. The environmental consequences of case-study decisions will be screened with the RF1 sustainability framework[21, p. 24].

4 Sustainability goals of the project and relation to the CEBE program vision

HAI-BRI contributes to CEBE vision of a built environment in harmony with planetary boundaries by addressing risk-informed life extension (Gap 7, RF7) based on timely and trustworthy

interpretation of deterioration. In bridges, much avoidable material use and CO₂ stems from late detection. Failing expansion joints and drainage let water and chlorides reach the reinforcement, and a local repair escalates into deck rehabilitation or replacement.

Objectives. (O1) *Least intervention*: **HAI-BRI** supports decisions (inspect, repair, replace) that select the minimum action consistent with safety. (O2) *Information sufficiency*: **HAI-BRI** identifies the minimum information needed for a trustworthy assessment, measurements and computation are added only where they change the decision.

Trade-offs and methodological choices. Digitalization is not sustainable per se. LLM/VLM inference consumes energy and resources. **HAI-BRI** therefore adopts a fit-for-purpose principle. Existing records are exploited before new data are acquired. Postponing action is sustainable only if deterioration is not underestimated. The separation of evidence, prior knowledge and inference, and engineer oversight are the safeguards against automation bias.

SDG perspective: Bridge infrastructures connect settlements and guaranty the delivery of goods and services. The rise in temperature and increase in rainfall negatively impact the corrosion mechanisms affecting common construction materials like concrete and steel. Therefore **HAI-BRI** SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action).

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